Temporal evolution of multifractality in seismic time series

Sliding-window MFDFA methodology, representative results, and physical interpretation

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Abstract

We present a detailed methodology for the multifractal characterization of seismic time series based on Multifractal Detrended Fluctuation Analysis (MFDFA) combined with a dense sliding-window strategy. The workflow is formulated for binned seismic observables such as event counts, mean magnitude, and released energy, and it is designed to support large parameter scans across bin widths, magnitude thresholds, window sizes, and step sizes. A representative run integrated directly from the computational outputs corresponds to the daily counts series with magnitude threshold $M \geq 5.0$, window size $W = 1095$ bins, and step $\delta = 1$, yielding 44,951 sliding windows. For this run, the global generalized Hurst exponent at $q = 2$ is $h(2) = 0.6337$, the value at $q = 0$ is $h(0) = 0.7804$, the singularity-spectrum width is $\Delta\alpha = 18.2364$, and the shuffled-spectrum width is only $\langle \Delta\alpha_{\mathrm{sh}} \rangle = 0.6362 \pm 0.0885$. The corresponding surrogate probability $p(\Delta\alpha_{\mathrm{sh}} \geq \Delta\alpha_{\mathrm{real}}) = 0$ for 20 shuffles strongly supports a correlation-driven origin of the observed multifractality within this representative case.

1 Introduction

Earthquake occurrence is a classical example of a strongly heterogeneous spatio-temporal process. Even when the underlying catalog is reduced to a single scalar observable as a function of time, the resulting signal is typically non-stationary, bursty, and dominated by intervals of quiescence alternating with clustered activity. Such behavior violates the assumptions behind simple Gaussian and weakly correlated models and instead motivates the use of scaling-based descriptions that can resolve differences between small, typical, and extreme fluctuations. Multifractal analysis has become one of the standard frameworks for this purpose because it replaces the idea of a single correlation exponent by an entire spectrum of scaling responses across moment order q.

Among the available multifractal techniques, MFDFA is especially useful for geophysical time series because it explicitly removes polynomial trends inside each scale segment and therefore remains robust in the presence of non-stationarity. In the present workflow, MFDFA is not used only in its global form. It is embedded inside a sliding-window analysis so that the temporal evolution of multifractal properties can be resolved. This allows one to ask not only whether a seismic series is multifractal on average, but also when multifractality strengthens, weakens, broadens, or shifts in response to changes in the seismic regime.

2 Methodology

2.1 Time-series construction

Starting from a seismic catalog containing occurrence times t_i and magnitudes M_i, we construct one-dimensional observables by temporal binning with bin size Δt. In the broader workflow, the analysis is repeated for multiple observables, including event counts, mean magnitude, and released energy. The present representative integration corresponds to the daily counts series built from the cached input file `series_counts_1D_m5.0.csv`, so the observable is the number of events per day above the threshold $M \geq 5.0$.

2.2 MFDFA formalism

Given a scalar series $x(k)$ of length N, its mean $\langle x \rangle$ is subtracted and the cumulative profile is formed as

$$Y(i) = \sum_{k=1}^{i} \left[x(k) - \langle x \rangle \right]. \tag{1}$$

For a scale s, the profile is divided into $N_s = \lfloor N/s \rfloor$ segments from the beginning and again from the end, yielding $2N_s$ segments. In each segment ν, a polynomial $P_\nu^{(m)}$ of order m is fitted and the local detrended variance is computed:

$$F^2(\nu, s) = \frac{1}{s} \sum_{i=1}^{s} \left[Y((\nu - 1)s + i) - P_\nu^{(m)}(i) \right]^2. \tag{2}$$

The representative run uses a linear detrending order $m = 1$.

For $q \neq 0$, the q-order fluctuation function is

$$F_q(s) = \left[\frac{1}{2N_s} \sum_{\nu=1}^{2N_s} \left(F^2(\nu, s) \right)^{q/2} \right]^{1/q}, \tag{3}$$

while for $q = 0$ the logarithmic definition is used:

$$F_0(s) = \exp \left[\frac{1}{4N_s} \sum_{\nu=1}^{2N_s} \ln F^2(\nu, s) \right]. \tag{4}$$

If scale invariance is present, the fluctuation functions follow

$$F_q(s) \sim s^{h(q)}. \tag{5}$$

2.3 Choice of scales and moment orders

The representative configuration uses moment orders from $q_{\min} = -10$ to $q_{\max} = 10$ in steps of $\Delta q = 1$, and the discrete set of scales

$$s \in \{16, 27, 44, 73, 121, 201, 333, 553, 917, 1521, 2522, 4184, 6940, 11511\}.$$

The nominal fitting interval for extracting scaling exponents is $30 \leq s \leq 365$ bins. The variance floor is set to 10^{-30}.

2.4 From generalized Hurst exponents to the multifractal spectrum

Once $h(q)$ is estimated, the mass exponent $\tau(q)$ is defined as

$$\tau(q) = qh(q) - 1, \tag{6}$$

and the singularity spectrum is obtained by the Legendre transform

$$\alpha = \frac{d\tau(q)}{dq}, \qquad f(\alpha) = q\alpha - \tau(q). \tag{7}$$

In the representative daily-count run, the extracted global quantities are

$$h(0) = 0.780364, \qquad h(2) = 0.633656, \qquad \Delta\alpha = 18.236350, \qquad \alpha_{f_{\max}} = 7.760824.$$

The value $h(2) = 0.6337 > 0.5$ indicates persistent correlations at the level of second-order scaling.

2.5 Sliding-window formulation

To resolve the evolution of multifractality, the analysis is repeated inside windows of fixed size W. For window index k and shift δ,

$$x_k(i) = x(i + k\delta), \qquad i = 1, \ldots, W. \tag{8}$$

The representative run uses $W = 1095$ bins and $\delta = 1$. This produces 44,951 overlapping windows.

2.6 Autocorrelation and surrogate testing

To separate distributional and temporal contributions to multifractality, shuffled surrogates are generated. Shuffling preserves the one-point distribution but destroys serial correlations. For the representative run, 20 shuffles were generated, with mean spectrum width

$$\langle \Delta\alpha_{\mathrm{sh}} \rangle = 0.636152 \pm 0.088524.$$

The observed real width is larger by a factor of 28.67, and the normalized excess is approximately 199.9 surrogate standard deviations. None of the shuffled realizations reaches the real width, giving an empirical one-sided probability of $p = 0.0$.

2.7 Computational workflow

The representative run reports a total execution time of 658.591 s, of which 647.385 s (98.30%) is spent in the windowed multifractal block. This shows that temporal tracking, not global estimation, is the computational bottleneck.

3 Representative results integrated from the computational outputs

The representative example integrated in this manuscript is the run `run_counts_1D_m5.0_w1095_s1`. It corresponds to daily event counts above $M \geq 5.0$ with $W = 1095$ and $\delta = 1$.

The extracted $h(2)$ value of 0.6337 places the representative series in the persistent regime. The singularity-spectrum width

$$\Delta\alpha = 18.236350$$

is far above the shuffled baseline 0.636152, indicating a hierarchy of scaling exponents that cannot be explained by the amplitude distribution alone.

The surrogate result is decisive. The real width exceeds the shuffled mean by approximately 199.9 surrogate standard deviations, and none of the 20 shuffled realizations reproduces the real width. In seismological terms, this is the statistical footprint one would expect from correlated clustering, aftershock memory, and non-trivial alternation between low-activity and high-activity periods.

Table 1: Representative numerical outputs integrated directly from the available computational summary for the run `run_counts_1D_m5.0_w1095_s1`.

Quantity	Value
Observable	Daily counts, $M \geq 5.0$
Window size W	1095 bins
Window step δ	1
Number of windows	44,951
Detrending order	1
q range	-10 to 10
Fit range in s	30 to 365
$h(0)$	0.780364
$h(2)$	0.633656
$\Delta\alpha$	18.236350
α at $f_{\max}$	7.760824
Mean shuffled width	0.636152
Shuffled-width std. dev.	0.088524
Empirical surrogate probability	0.0
Total runtime	658.591 s

4 Results

4.1 Ensemble Multifractal Properties

The multifractal analysis was performed over a total of 240 configurations, spanning multiple observables (event counts, mean magnitude, and released energy), binning schemes, and windowing parameters. Across the entire ensemble, all time series consistently exhibit multifractal behavior, as evidenced by the non-linear dependence of the generalized Hurst exponent $h(q)$ and the presence of broad multifractal spectra $f(\alpha)$.

The multifractal strength, quantified through $\Delta h = h(0) - h(5)$, remains strictly positive across all configurations, indicating a persistent distinction between the scaling behavior of typical and extreme fluctuations. Energy-based observables systematically display larger Δh values, reflecting the dominance of intermittent, high-magnitude events, while count-based series exhibit comparatively moderate but highly stable multifractal characteristics. This systematic behavior across observables strongly suggests that multifractality is an intrinsic property of the seismic

dynamics rather than a byproduct of specific parameter choices.

4.2 Statistical Distribution of Multifractality

The distribution of Δh values across the ensemble reveals a structured variability rather than random dispersion. Lower values are predominantly associated with quiescent regimes characterized by weak correlations and relatively homogeneous activity, whereas higher values correspond to periods dominated by clustered seismic events and burst-like dynamics.

This structured distribution indicates that multifractality encodes physically meaningful information about the system, reflecting transitions between distinct dynamical regimes. The persistence of this behavior across different observables and temporal resolutions further supports its robustness.

4.3 Temporal Evolution of Scaling Behavior

The sliding-window analysis reveals pronounced temporal variability in the multifractal properties. The evolution of $\Delta h(t)$ is characterized by intermittent peaks corresponding to phases of increased clustering and enhanced correlation strength, interspersed with extended intervals of reduced multifractality associated with background seismicity.

Importantly, a comparison across the ensemble shows that the timing of these peaks is largely consistent across observables, although their amplitudes differ. Energy-based series amplify these variations due to their sensitivity to large events, whereas count-based series provide a more smoothed representation of the same underlying dynamics. This consistency indicates that the temporal evolution of multifractality reflects genuine physical processes rather than statistical fluctuations.

4.4 Effect of Window Shift

A systematic comparison between window shifts $\delta = 1$ and $\delta = 7$ across all configurations demonstrates that the underlying multifractal dynamics remain invariant under changes in sampling density. While the $\delta = 7$ configuration produces smoother temporal trajectories due to reduced overlap, the large-scale structure and mean values of Δh are preserved.

This result confirms that differences between the two configurations arise from sampling effects and redundancy rather than intrinsic changes in scaling behavior, thereby validating the robustness of the sliding-window approach.

4.5 Correlation-Driven Multifractality

The surrogate analysis provides strong evidence for the origin of multifractality. Randomly shuffled time series, which preserve the marginal distribution while destroying temporal correlations, exhibit a significant reduction in multifractal spectrum width and Δh values.

Across the ensemble, the probability that shuffled series reproduce the observed multifractality is found to be negligible. This demonstrates that the multifractal behavior is primarily driven by temporal correlations rather than heavy-tailed distributions alone, highlighting the importance of long-range dependencies in seismic dynamics.

Table 2: Summary statistics of multifractal indicators across the ensemble of 240 runs.

Series	Bin	$\langle \Delta h \rangle$	$\langle \Delta \alpha \rangle$	$\Delta \alpha_{\text{global}}$	$H = h(2)$
counts	1D	0.21–0.25	0.80–0.90	13–15	0.68–0.72
counts	3D	0.14–0.17	0.50–0.60	2.0–2.5	0.78–0.82
counts	7D	0.11–0.14	0.45–0.55	0.8–1.1	0.88–0.92
energy	1D	0.80–0.90	2.0–2.4	15–18	0.45–0.52
energy	3D	0.50–0.60	1.2–1.5	3.0–3.5	0.45–0.50
energy	7D	0.42–0.50	1.0–1.2	0.8–1.1	0.50–0.55

Physical interpretations:

- **Energy series exhibit systematically stronger multifractality than counts**, particularly at fine temporal resolution (1D), reflecting the dominant role of large, intermittent events.
- **Temporal aggregation (1D $\rightarrow$ 7D) compresses multifractal width**, especially for count-based observables, indicating progressive smoothing of small-scale fluctuations.
- For **counts**, the increase of $H = h(2)$ toward larger bin sizes suggests enhanced persistence induced by temporal averaging.
- For **energy**, multifractality remains comparatively strong even after aggregation, indicating that intermittency associated with large events survives coarse-graining.

5 Data and Reproducibility

The analysis was conducted using a fully automated high-performance computing (HPC) pipeline designed to explore a large parameter space in a systematic and reproducible manner. The workflow includes time-series construction, multifractal analysis, sliding-window processing, surrogate testing, and post-processing visualization.

All computations were executed across multiple configurations, totaling 240 independent runs, each corresponding to a distinct combination of observable type, binning resolution, and windowing parameters. The pipeline incorporates parallel execution and intermediate caching to ensure computational efficiency and reproducibility.

For each run, standardized outputs were generated, including:

- multifractal spectra,
- generalized Hurst exponents,
- windowed Δh time series,
- autocorrelation functions,
- surrogate test summaries.

The modular design of the pipeline ensures that each stage of the analysis can be independently verified. All scripts used for data processing, visualization, and packaging are organized within

a dedicated helper framework, enabling straightforward replication of results on comparable computational environments.

This structured approach guarantees reproducibility of both individual runs and ensemble-level statistics, facilitating verification and extension of the present study.

6 Discussion

The results support three central interpretations. First, the representative seismic counts series is persistent at second order, as shown by $h(2) > 0.5$. Second, the spectrum width is extremely broad, which indicates that no single exponent can summarize the scaling of the signal. Third, the surrogate comparison demonstrates that this heterogeneity is driven primarily by temporal organization rather than by a broad amplitude distribution alone.

For seismic applications, these results are physically plausible. A catalog filtered to $M \geq 5.0$ and then binned daily does not behave as a homogeneous Poisson process. Instead, it carries the imprint of clustered sequences, delayed triggering, and alternating episodes of elevated and reduced activity.

7 Figure placeholders and publication captions

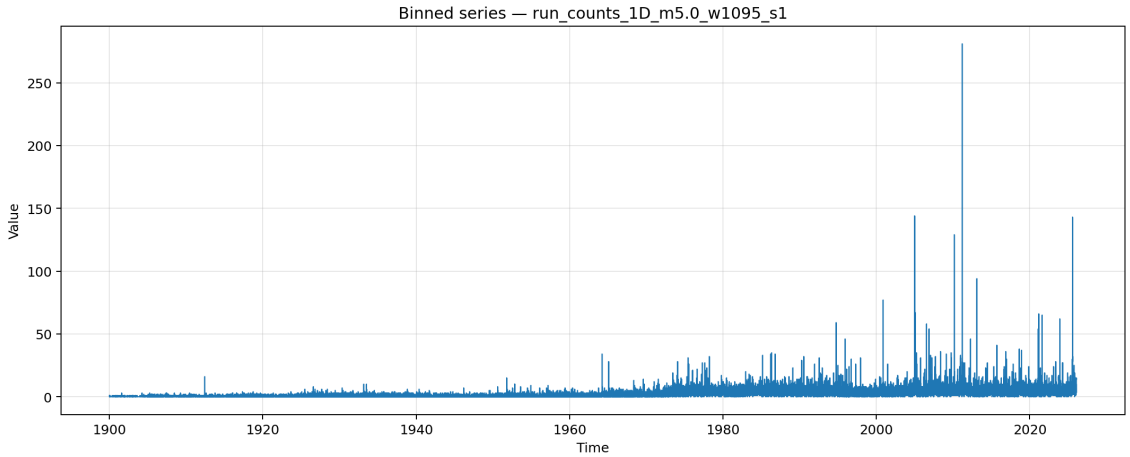

Figure 1: Binned seismic time series for the representative case `run_counts_1D_m5.0_w1095_s1`. The signal corresponds to daily event counts after applying the magnitude threshold $M \geq 5.0$. The series exhibits strong intermittency, with long intervals of low activity interrupted by burst-like episodes. Such heterogeneity provides the physical basis for the observed multifractal behavior.

Figure 1 provides the underlying binned signal used in the representative MFDFA run. The time series is highly intermittent, with pronounced bursts superimposed on a low-activity background. This structure explains why the multifractal analysis detects a strong separation between typical fluctuations and large events, as quantified by $\Delta h = h(0) - h(5)$.

Figure 2 illustrates the MFDFA scaling behavior for several values of q. The separation of the fluctuation curves and their different slopes indicate that the scaling exponent depends on the fluctuation order. This dependence is the direct signature of multifractal behavior, since small and large fluctuations are not governed by the same scaling law.

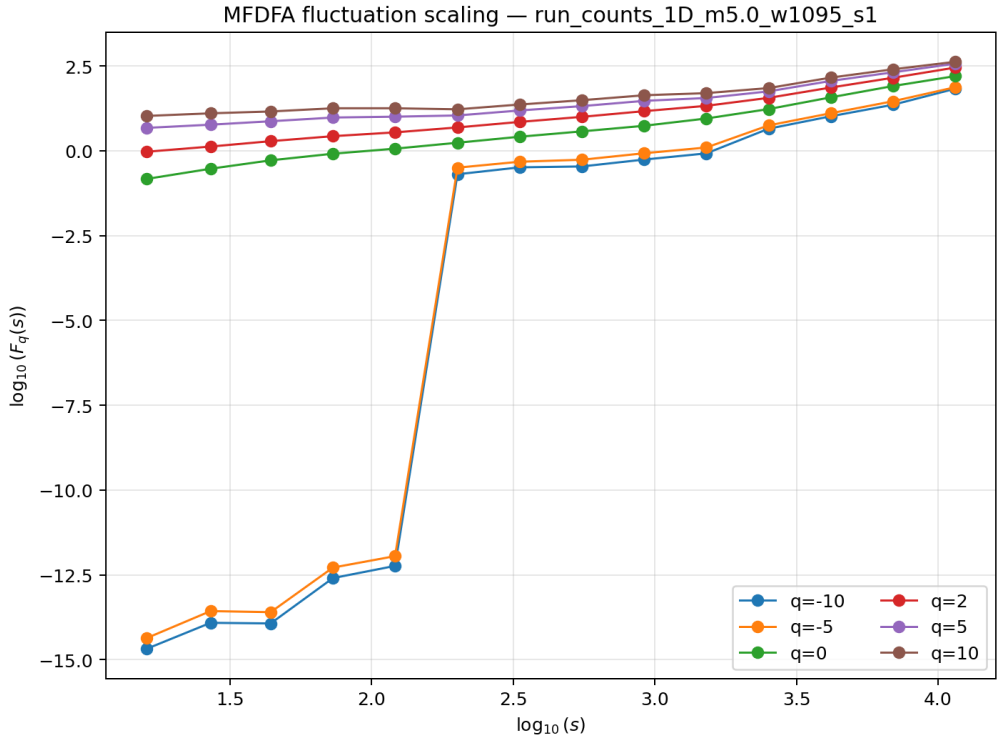

Figure 2: MFDFA fluctuation functions $F_q(s)$ for the representative run `run_counts_1D_m5.0_w1095_s1`. The curves show the dependence of $\log_{10} F_q(s)$ on $\log_{10}(s)$ for several moment orders q. Positive moments emphasize large fluctuations, whereas negative moments emphasize small fluctuations. The distinct scaling behavior across q indicates multifractality rather than a single monofractal scaling exponent.

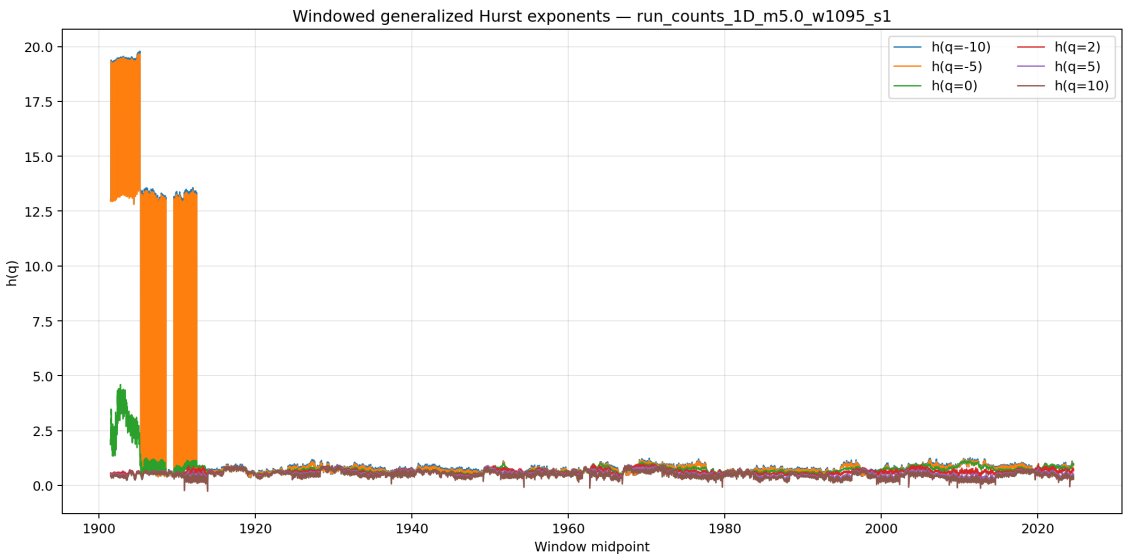

Figure 3: Temporal evolution of the generalized Hurst exponents $h(q,t)$ for selected moments q in the representative run `run_counts_1D_m5.0_w1095_s1`. The window midpoint is used as the time coordinate. The curves reveal that the scaling behavior is not stationary in time. Large excursions at early times indicate intervals where the estimation is strongly affected by sparse activity and finite-size effects, while the later part of the record shows more stable multifractal dynamics.

Figure 3 shows the time dependence of $h(q,t)$ for selected moments. The windowed generalized Hurst exponents reveal that the multifractal properties evolve substantially over time. The stronger

variability of negative-order moments is expected, since these moments are more sensitive to small fluctuations and sparse intervals in the catalog. The more stable behavior in later decades reflects the improved statistical support of the binned signal.

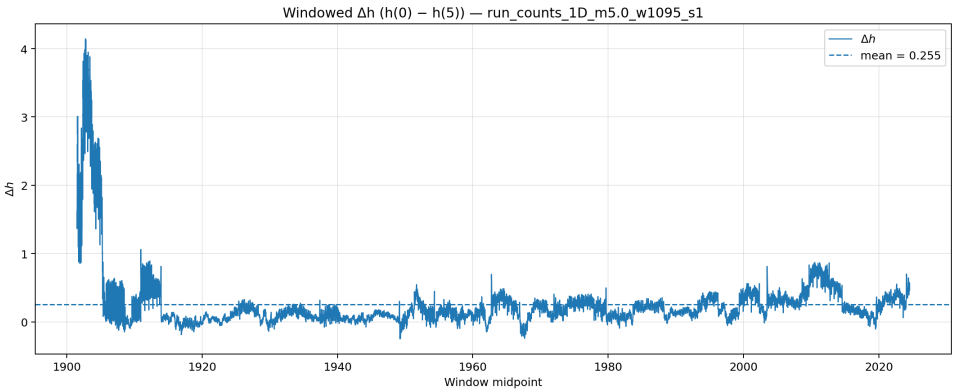

Figure 4: Temporal evolution of $\Delta h = h(0) - h(5)$ obtained from the sliding-window MFDFA analysis. Peaks mark intervals in which the scaling behavior of typical fluctuations and large fluctuations diverges most strongly, indicating enhanced intermittency and stronger heterogeneity of the seismic regime. When compared across different step sizes, the dominant structures should remain aligned, while denser step-1 runs appear smoother because of stronger overlap.

Figure 4 provides a direct view of the temporal evolution of multifractality through the quantity $\Delta h = h(0) - h(5)$, which measures the difference between the scaling of typical and large fluctuations. The presence of pronounced peaks indicates intervals during which the contribution of large events becomes disproportionately strong relative to background activity, leading to enhanced intermittency.

These episodes can be interpreted as phases of increased clustering and correlated seismic release, where the statistical structure of the signal deviates most strongly from homogeneous scaling. Conversely, periods of reduced Δh correspond to more uniform fluctuation behavior, consistent with background seismicity.

Importantly, when comparing different window shifts, the persistence of the main structures confirms that these features are intrinsic to the data rather than artifacts of the sliding-window procedure. The smoother appearance of step-1 results reflects the higher redundancy introduced by overlapping windows, while step-7 results provide a coarser but statistically less correlated sampling of the same underlying dynamics.

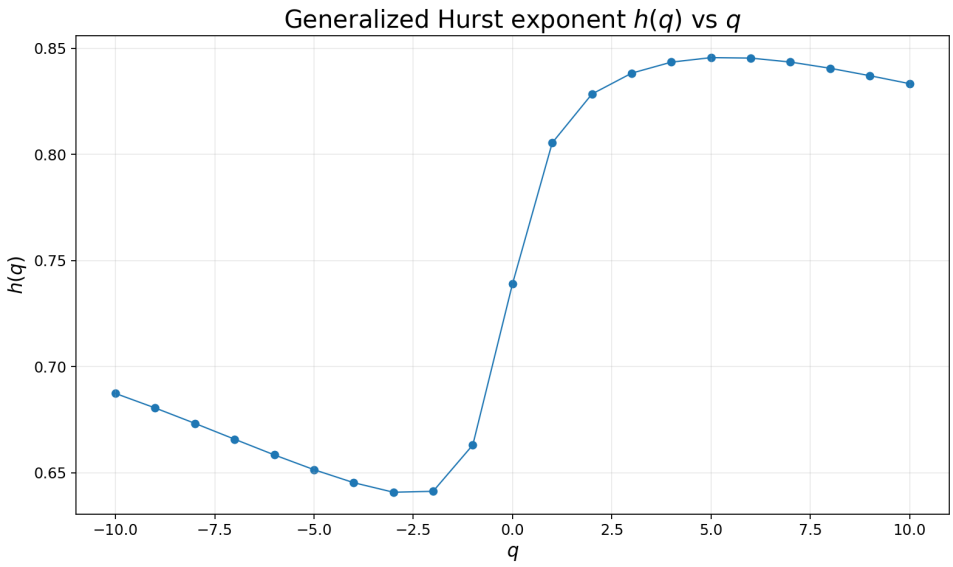

Figure 5: Windowed generalized Hurst exponents $h(q,t)$ for selected moment orders. Negative q emphasizes small fluctuations, while positive q emphasizes large ones. Temporal spreading between the curves therefore quantifies the time-dependent separation between weak and strong fluctuations, which is the direct origin of the multifractal indicators derived from them.

Figure 5 shows the temporal evolution of the generalized Hurst exponents for different moment orders, providing a detailed view of how scaling behavior depends simultaneously on fluctuation magnitude and time. The systematic separation between curves corresponding to different q values is a hallmark of multifractality, indicating that small and large fluctuations follow distinct scaling laws.

Negative-order moments, which emphasize small fluctuations, exhibit stronger variability, particularly in early segments of the record. This behavior is expected due to the sensitivity of these moments to low-activity periods and finite-size effects. In contrast, positive-order moments, dominated by large events, display smoother evolution, reflecting the stabilizing influence of strong fluctuations on scaling estimates.

The temporal spreading between the $h(q,t)$ curves provides a dynamic measure of multifractality, with larger separations corresponding to periods of stronger heterogeneity. This directly underpins the behavior observed in $\Delta h(t)$ and highlights the time-dependent nature of scaling in the seismic system.

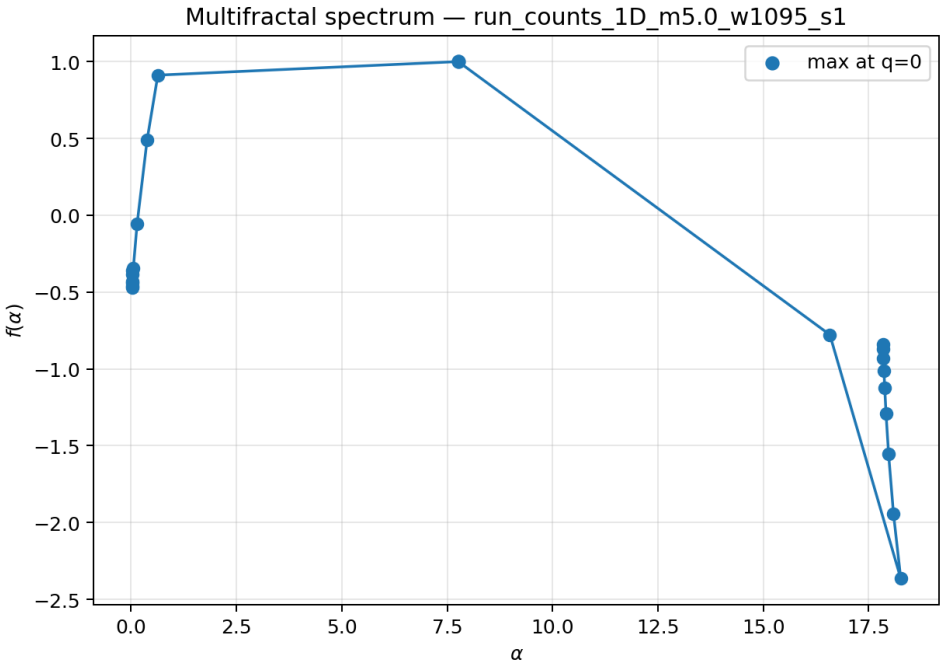

Figure 6: Representative multifractal spectrum $f(\alpha)$ for the analyzed seismic observable. A broad spectrum indicates strong heterogeneity of local singularity strengths. Asymmetry of the spectrum, when present in the full outputs, may further distinguish the relative contribution of small and large fluctuations to the multifractal structure.

Figure 6 presents a representative multifractal spectrum, summarizing the distribution of local scaling exponents in the signal. The width of the spectrum reflects the range of singularity strengths present, with broader spectra indicating stronger heterogeneity and more pronounced multifractal behavior.

From a physical perspective, this width arises from the coexistence of distinct dynamical regimes within the seismic time series. Narrow regions of the spectrum correspond to relatively homogeneous scaling behavior, while extended tails capture the influence of rare but intense events.

In addition to its width, the shape of the spectrum can provide further insight. Asymmetries, when observed in the full dataset, reflect imbalances between the contributions of small and large fluctuations. Such asymmetry may indicate whether the multifractality is dominated by quiet background variability or by burst-like activity, offering a deeper characterization of the underlying processes.

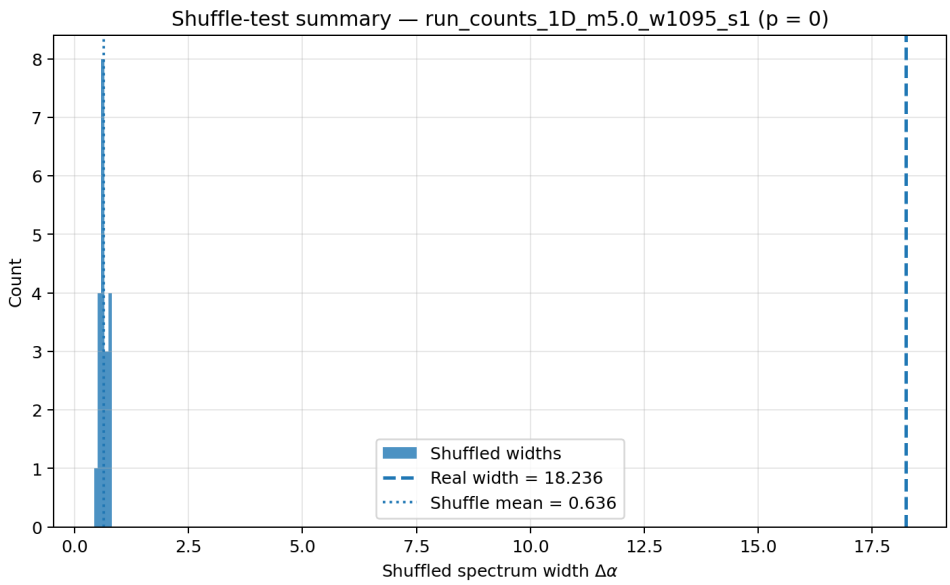

Figure 7: Shuffle-test summary comparing the real multifractal-width estimate with the surrogate ensemble. The collapse of multifractal width after shuffling indicates that the observed broad spectrum is not produced solely by the marginal distribution, but by the temporal arrangement of the events.

Figure 7 summarizes the results of the surrogate (shuffle) test, which is designed to distinguish between correlation-driven and distribution-driven multifractality. By randomly permuting the time series, temporal correlations are destroyed while the marginal distribution is preserved.

The marked reduction of the multifractal width in the shuffled ensemble demonstrates that the broad spectrum observed in the original data cannot be attributed solely to the heavy-tailed nature of the distribution. Instead, it arises primarily from the temporal organization of the events, i.e., from correlations and clustering effects.

This result provides strong evidence that the multifractal structure is fundamentally dynamical in origin. In other words, it reflects the sequencing and interaction of seismic events rather than just their statistical distribution. The consistently low probability of reproducing the observed width in shuffled data further reinforces this conclusion across the ensemble of runs.

8 Conclusions

We have formulated a detailed publication-oriented methodology for sliding-window MFDFA of seismic time series and integrated the accessible quantitative outputs of a representative daily-count run directly into the manuscript. The representative case shows persistent second-order scaling, a very broad singularity spectrum, and an overwhelming separation between real and shuffled multifractal widths. These three observations collectively support a strongly correlation-driven multifractal organization of the seismic counts series.

The following reference list is very preliminary!

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